

MLFF Architecture Revision 85

CUEQ-REPEAT1-DIAG3 warm-up + all-pairs repeatability refinement

mdstats project

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Revision 85

Release: mdstats 0.20.218a0

Dependency-graph schema: 67

Gate: CUEQ-REPEAT1-DIAG3

The revision-84 MPA-0/default workstation evidence confirmed that TRAIN2 FP32 force differences are dominated by a stochastic execution envelope rather than a stable e3nn/pure-CuEq offset. In ordinary execution, e3nn-self Fmax reached $1.962e-5$, CuEq-self $1.836e-5$, and cross-backend $2.270e-5$. Under the isolated deterministic controls, e3nn-self comparisons against run 1 became a constant $1.061e-5$, while CuEq-self still reached $2.709e-5$ and cross-backend $1.550e-5$. Selection remained identical throughout.

The constant deterministic e3nn self value exposed a second problem with the temporary diagnostic: every self comparison used run 1 as its reference, so a first-call/warm-up shift could contaminate all nine reported self comparisons. Revision 85 removes that arbitrary baseline.

Each backend now receives one complete **discarded warm-up evaluation** on the exact doctor probe. Ten post-warm-up outputs are then retained for e3nn and ten for pure CuEq. No additional model calls are needed for pair construction. mdstats computes offline:

- $C(10, 2) = 45$ e3nn-self pairs;
- $C(10, 2) = 45$ CuEq-self pairs;
- $10 \times 10 = 100$ e3nn/CuEq cross-backend pairs.

Every pair records Fmax, force RMSE, force absolute-error p99 and p99.9, the number of force components above $1e-5$, energy/stress/descriptor maximum errors, and selection identity where applicable. Terminal summaries report min, median, p90, p99, and max instead of printing all 100 cross pairs.

The ordinary diagnostic record advances to mdstats.training-acceleration-repeatability-diagnostic.v2. Historical v1 records remain readable and are interpreted using their original baseline-comparison semantics. The isolated deterministic-control subprocess remains schema v1 at the outer wrapper but carries the new v2 repeatability record when completed.

Revision 85 remains **non-authorizing**. TRAIN2 FP32 parity stays at $rtol=1e-5$, $atol=1e-5$; source/DATA6 and FP64 tolerances are unchanged. FINAL-GPU1 remains archival until the DIAG3 workstation distributions are reviewed and a permanent noise-normalized backend-equivalence criterion is frozen.